

Integration by Parts

Today's Goals

- ◆ You should be able to explain the process of integration by parts in relation to the Product Rule.
- ◆ You should be able to use integration by parts to evaluate definite and indefinite integrals.
- ◆ You should be able to determine whether integration by parts or integration by substitution is helpful in evaluating an integral.

1. Calculate $\int x \cdot \sin x^2 dx$. Can we use substitution to calculate $\int x \cdot \sin x dx$?

2. Integration by substitution allowed us to “undo” the chain rule. In this problem you are going to learn how to “undo” the product rule.

(a) State the product rule, in both functional and Leibniz notation:

$$\begin{aligned} \frac{d}{dx}(u(x) \cdot v(x)) &= \underline{\hspace{2cm}} \\ &= \underline{\hspace{2cm}} \end{aligned}$$

(b) Note that u and v are both functions of x . Now, integrate both sides of the functional notation version above with respect to x , and rearrange to find a formula for $\int u(x)v'(x)dx$. Hint: Recall the Fundamental Theorem of Calculus I!

Similarly, take the integral with respect to x of both sides of the Leibniz version and rearrange to find $\int u \frac{dv}{dx} dx$.

Integration by Parts

If $u(x), v(x)$ are both differentiable functions of x , then

$$\int u(x) \cdot v'(x) dx = \underline{\hspace{2cm}}$$

We often write $u'(x)dx = du$ and $v'(x)dx = dv$, in which case the formula is

$$\int u dv = \underline{\hspace{2cm}}$$

3. In this problem you will calculate $\int x \cdot \sin x dx$.

- (a) If we want to calculate $\int x \cdot \sin x dx$ using the technique of integration by parts, we must interpret it as an integral of the form $\int u(x) \cdot v'(x) dx$. What should we make u and what should make $\frac{dv}{dx}$? Is this the only choice?

- (b) Use the formula for integration by parts and your choices for u and v to calculate $\int x \cdot \sin x dx$.

4. Evaluate the following integrals, using integration by parts. Clearly state what you are choosing for u and what you are choosing for v .

(a) $\int x \cdot \ln x dx$

(b) $\int x e^x dx$

(c) $\int_0^1 x e^x dx$

(d) $\int x^2 e^x dx$

(e) $\int \ln x dx$

(f) $\int_0^{\ln 3} x^2 e^{2x} dx$

5. Evaluate the following integrals. Note: different techniques apply!

(a) $\int \frac{1}{0.5x - 3} dx.$

(b) $\int_0^{\sqrt{15}} 5x(x^2 + 1)^{-9/4} dx.$

(c) $\int \left(\sqrt[3]{x} + \frac{1}{\sqrt[3]{x^2}} \right)^3 dx.$

(d) $\int \frac{4x + 4}{2x + 1} dx.$

(e) $\int \frac{x+2}{x^2+4x-5} dx.$

(f) $\int_{-10}^{10} \sin(x^3) dx.$

(g) $\int x \arctan x dx.$

(h) $\int x^2 \cos x dx.$